

Nishant Singh

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LinkedIn

PROFESSIONAL SUMMARY

Backend and Infrastructure Automation Engineer with **2 years of experience** building **Python-based automation platforms** for **Cisco SD-WAN environments** at **Tata Communications**. Experienced in developing scalable backend utilities, infrastructure automation workflows, validation systems, and upgrade orchestration tools using **Python, FastAPI, Flask, REST APIs, Playwright, Paramiko, PostgreSQL, and Linux**.

Built automation solutions that significantly reduced manual operational effort, improved deployment consistency, accelerated validation workflows, and streamlined SD-WAN infrastructure management. Currently strengthening expertise in **Backend Engineering, Data Structures & Algorithms, System Design, DevOps, and Scalable Distributed Systems**.

TECHNICAL SKILLS

Programming Languages	Python
Backend Development & APIs	FastAPI, Flask, REST APIs, JSON, Async Programming
Automation & Testing	Playwright, Selenium, Paramiko
Databases	MySQL, PostgreSQL
Cloud, DevOps & Containers	Docker, Linux, GitHub Actions, Jenkins
Security & API Management	JWT Authentication (Learning), API Security (Basics)
Version Control & Project Management	Git, GitHub, Jira
Networking & Infrastructure	Cisco SD-WAN, TCP/IP, DNS, HTTP/HTTPS, Routing & Switching
Monitoring & Troubleshooting	Wireshark, tcpdump

EXPERIENCE

Engineer – Cisco SD-WAN & Infrastructure Automation

July 2024 – Present

Tata Communications Limited, Chennai, India

- Developed **Python-based automation frameworks** for Cisco SD-WAN operational workflows, reducing repetitive manual effort and improving deployment efficiency across validation environments.
- Built backend automation utilities using **Flask, REST APIs, Paramiko, and Playwright** to automate **20+ operational workflows** including software upgrade orchestration, infrastructure validation, and centralized log collection.
- Integrated **Cisco vManage REST APIs** for real-time device validation, infrastructure management, and operational automation on **multiple SD-WAN controllers and edge devices**.
- Designed and implemented **pre/post-upgrade validation workflow** to automate health checks, improve deployment reliability, and ensure configuration consistency.
- Developed web-based automation interfaces with **real-time execution monitoring and live log streaming** capabilities for operational visibility.
- Automated repetitive operational tasks, reducing the manual troubleshooting and backup effort by approximately **65%** and improving the operational response time.
- Worked extensively with **Linux-based environments**, Git/GitHub workflows, and infrastructure troubleshooting during automation development and testing.
- Applied networking concepts including **TCP/IP, DNS, HTTP/HTTPS, DHCP, NAT, routing, and SD-WAN architecture** during automation and operational troubleshooting.

PROJECTS

Cisco SD-WAN Platform Automation

- Developed **Python-based automation frameworks** for Cisco SD-WAN operational workflows, reducing repetitive manual effort by **60%** and improving deployment efficiency in validation environments.
- Implemented automated upgrade execution, version management, centralized log collection, and **20+ pre/post validation checks**, reducing upgrade validation time from **2–3 hours to under 10 minutes**.
- Designed web-based interfaces for automation execution, workflow visibility, and live operational log streaming.

Configuration Backup & Infrastructure Validation Automation

- Developed Python-based automation script using **Paramiko** to automate CLI login, configuration backup, and infrastructure validation across SD-WAN devices and controllers.
- Automated configuration backup workflows, reducing backup and collection time from manual execution to approximately **5 minutes**.
- Implemented centralized operational log collection and version-controlled backup workflows using **Git/GitHub** to improve recovery reliability and operational traceability.
- Reduced repetitive operational effort and accelerated troubleshooting workflows through automated infrastructure backup and validation processes.

CERTIFICATIONS

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| • Cisco Catalyst SD-WAN Foundations – LinkedIn Learning | Oct 2024 |
| • Introduction to SD-WAN – LinkedIn Learning | Oct 2024 |
| • Meta Front-End Developer Professional Certificate – Meta | Jan 2024 |
| • AWS Academy Graduate - AWS Academy Cloud Foundation – Amazon Web Services (AWS) | May 2023 |
| • AWS Academy Graduate - AWS Academy Cloud Architecting – Amazon Web Services (AWS) | May 2023 |
| • AWS Academy Graduate - Engineering Operations Technicians – Amazon Web Services (AWS) | Nov 2022 |
| • Complete Web Development Bootcamp – Udemy | Aug 2022 |

EDUCATION

Bachelor of Technology in Computer Science and Engineering

S.R.M. Institute of Science and Technology

2020 – 2024

CGPA: **8.88 / 10**